

Question_1:

DS	Session	Arrays	Question Information	Level 1	Challenge 21
Problem Description: How many Y's did a Roman Centurion make a day in cold hard Lira? About a C's worth! Turns out, Martians gave Rome the idea for their number system. Use the conversion charts below to help translate some Martian numbers! Note, that unlike the Roman Numerals, Martian Numerals reuse symbols to mean different values. B can either mean '1' or '100' depending on where it appears in the number sequence. Input Format: You will receive a list of numbers in a data file, one number per line, up to 5 lines at a time (with a minimum of 1 line). No number will exceed 100 or be less than 1. Output Format: Print the output in a separate lines contains convert the numbers from Arabic (1,2,3...10...500...1000) to Martian (B,BB,BBB...Z...G...R) numerals.					

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
26 67	161 627
186 408 500	816 401 501
EXPECTED OUTPUT	EXPECTED OUTPUT
ZZWB PZWBB BPZZZWB BGWBBB G	BPZB GBZZWBB GBBBZWB BGB GB

Mandatory Test Cases

Test Case 1	Test Case 2	Test Case 3
KEYWORD	KEYWORD	KEYWORD
char buf[]	buf[i++]='R';	while(n>=10)

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
18	455	84

Code:

```
1  /*
2  Question_1:
3  Convert each Arabic decimal number from 1 to 1000 into its
4  corresponding Martian numeral and print one result per line.
5  */
6
7  #include <iostream>
8  #include <string>
9  using namespace std;
10
11 int main()
12 {
13     string ones[] =
14     {
15         "", "B", "BB", "BBB", "BW",
16         "W", "WB", "WBB", "WBBB", "BZ"
17     };
18     string tens[] =
19     {
20         "", "Z", "ZZ", "ZZZ", "ZP",
21         "P", "PZ", "PZZ", "PZZZ", "ZB"
22     };
23     string hundreds[] =
24     {
25         "", "B", "BB", "BBB", "BG",
26         "G", "GB", "GBB", "GBBB", "BR"
27     };
28     int n;
29
30     while (cin >> n)
31     {
32         char buf[50];
33         int i = 0;
34
35         if (n == 1000)
36         {
37             buf[i++] = 'R';
38         }
```

```
39     else
40     {
41         int h = n/100;
42         n %= 100;
43
44         int t = n/10;
45         n %= 10;
46
47         int o = n;
48
49         string result = "";
50
51         result += hundreds[h];
52         result += tens[t];
53         result += ones[o];
54
55         for (char ch : result)
56         {
57             buf[i++] = ch;
58         }
59     }
60     buf[i] = '\0';
61     cout << buf << endl;
62 }
63 return 0;
64 }
```

Output:

Test Case_1:

```
26 67 186 408 500
ZZWB
PZWBB
BPZZZWB
BGWBBB
G
```

Test Case_2:

```
161 627 816 401 501
BPZB
GBZZWBB
GBBBZWB
BGB
GB
```